TEAM ACTIVITY (WEEK 04)

1) Program Foundation #1: Abstraction with YouTube Videos

1.1 What does the program do?

The program models YouTube videos by recording information such as title, author, and duration, and manages the comments associated with each video, storing the author and text of each comment. All videos are saved in a collection that the program browses to display complete data for each video along with its comments.

1.2 List of Candidate classes

- Video
- Comment
- Program (The one who organizes everything/entry point)

1.3 Responsibilities per class

Video class

- Represent a YouTube video.
- Maintain its basic information.
- Manage its comment list.
- Provide the comment count.

Comment class

- Represent an individual comment.
- Store the author and comment text.

Program class

- Create instances of Video and Comment.
- Populate test data.
- Display the requested information.

1.4 Class Design

Comment Class

Attributes (private)

- string _author
- string _text

Méthods

- Comment(string author, string text) (constructor)
- string GetAuthor()
- string GetText()

Video Class

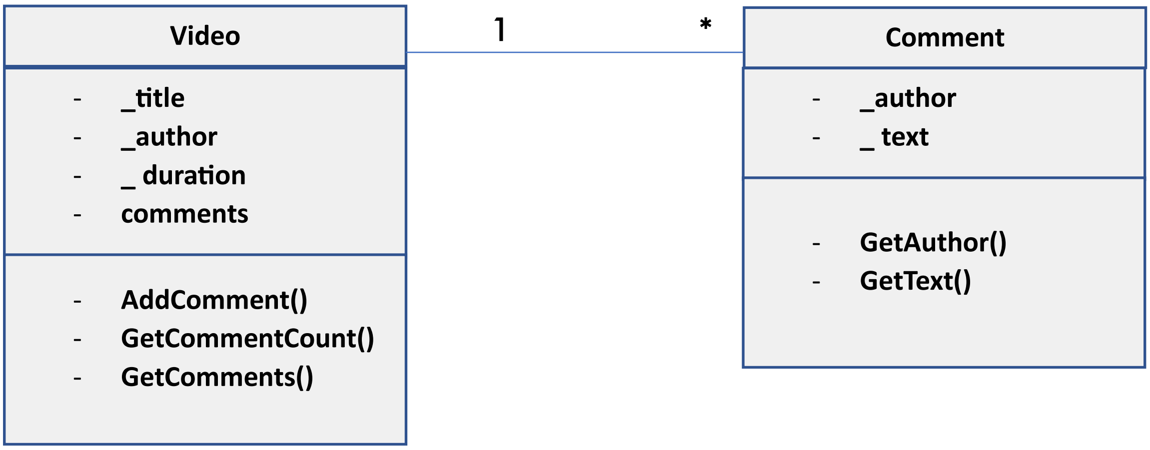
Attributes (private)

- string _title
- string _author
- int _durationSeconds
- List<Comment> _comments

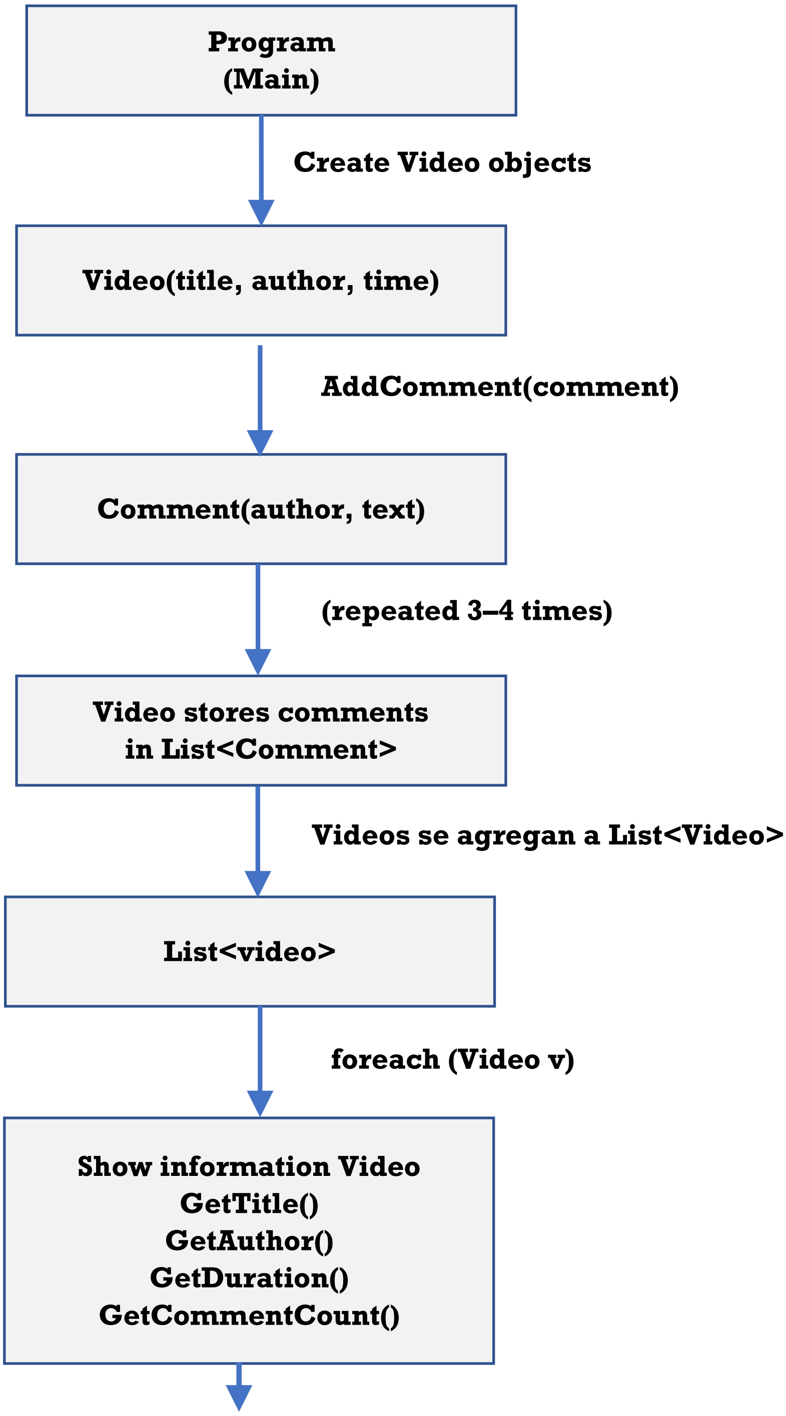
Méthods

- Video(string title, string author, int durationSeconds)
- void AddComment(Comment comment)
- int GetCommentCount()
- string GetTitle()
- string GetAuthor()
- int GetDuration()
- List<Comment> GetComments()

1.5 Class Diagram



1.6 How the program will run or how these methods relate to one another.



GetComments()

Show information Video
GetAuthor()
GetText()

<u>Program description</u>

The Main method will be the program's integrator.
The program will create Video objects and add Comment objects to them using the AddComment() method.
Each Video object will encapsulate its list of comments (List<Comment>).
All videos will be stored in the collection (List<Video>).
The program will iterate through this collection and, for each video:
• obtain the basic data using methods (GetTitle, GetAuthor, etc.),
• obtain the number of comments with GetCommentCount(),
• and finally iterate through the comments using GetComments().
For each video, the following information is printed:

- Title
- Author
- Duration
- Number of comments
- List of comments

2) Foundation Program #2: Encapsulation with Online Orders

2.1 What does the program do?

It manages product orders, calculates the total cost (including shipping), and generates packaging and shipping labels, correctly encapsulating the data.

2.2 Candidates for classes

- Product
- Customer
- Address
- Order
- Program (orquestación)

2.3 Responsibilities by class

Product Class

- Represent a product from the order.
- Calculate its total cost.

Customer Class

- Represent a client.
- Provide residency information (U.S. or non-U.S.) via address.

Address Class

- Representar la dirección física.
- Determinar si está en EE. UU.
- Devolver la dirección formateada.

Order Class

- Contener productos y un cliente.
- Calcular el precio total del pedido.
- Generar etiquetas de embalaje y envío.

2.4 Class Design

Address Class

Attributes (private)

- string _street
- string _city
- string _stateOrProvince
- string _country

Methods

- Address(string street, string city, string stateOrProvince, string country)
- bool IsInUSA()
- string GetFullAddress()

Customer Class

Attributes (private)

- string _name
- Address _address

Methods

- Customer(string name, Address address)
- string GetName()
- bool LivesInUSA()
- string GetShippingAddress()

Product Class

Attributes (private)

- string _name
- string _productId
- decimal _price
- int _quantity

Methods

- Product(string name, string productId, decimal price, int quantity)
- decimal GetTotalCost()
- string GetName()
- string GetProductId()

Order Class

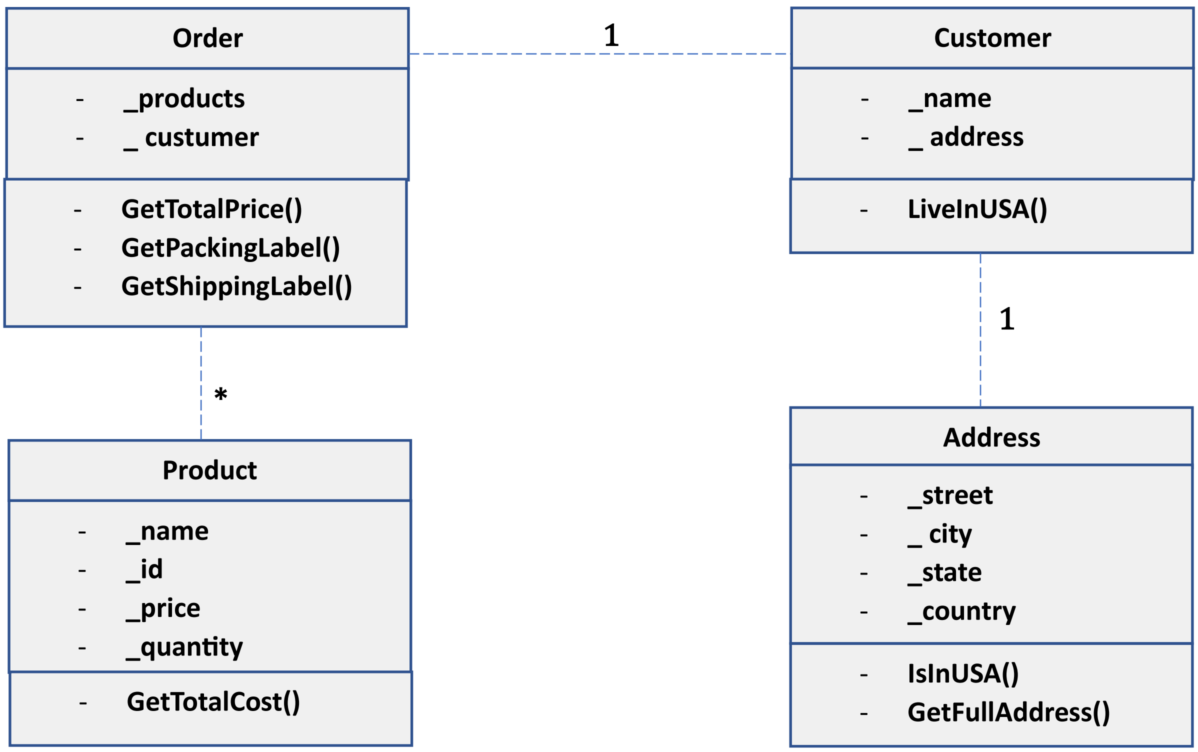
Attributes (private)

- List<Product> _products
- Customer _customer

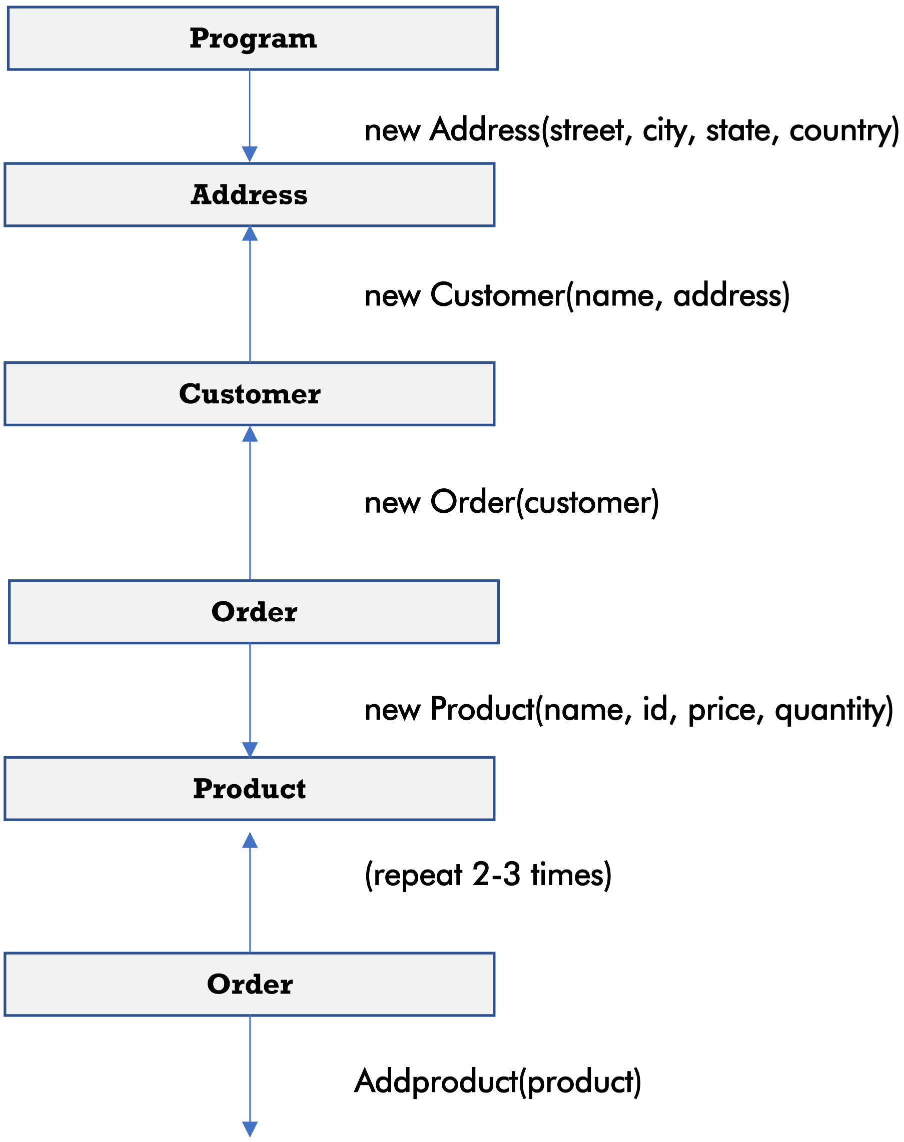
Methods

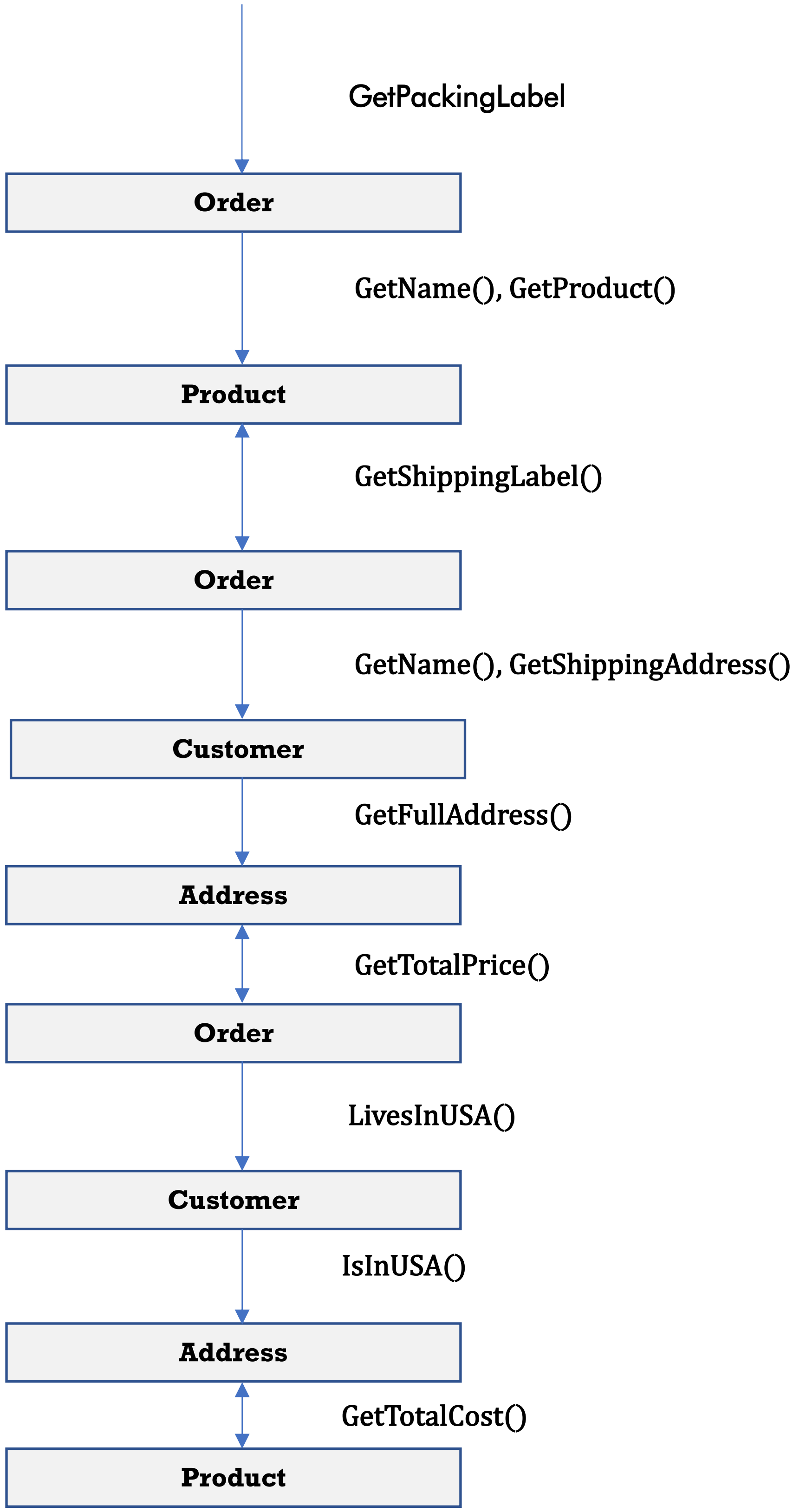
- Order(Customer customer)
- void AddProduct(Product product)
- decimal GetTotalPrice()
- string GetPackingLabel()
- string GetShippingLabel()

2.5 Class Diagram



2.6 How the program will run or how these methods relate to one another.





The program starts in Program, which coordinates execution by creating addresses and customers, then products and orders, to which these products are added. The Order class centralizes the order logic and, for each order, generates the packing label, shipping label, and total price, delegating specific responsibilities: Product calculates its own cost (`GetTotalCost()`), Address determines if the address is in the US (`IsInUSA()`), and Customer delegates this verification to its address.

For each order, the following are displayed:

- Packaging label.
- Shipping label.
- Total order price.